

## **Experiment Name: Designing a Class Diagram**

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### **Case Study: Career Guidance and Examination Portal**

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#### **Introduction:**

The Career Guidance Portal is an online system that helps students prepare for competitive exams and receive counselling. It also involves administrators, counsellors, advertisers, and test maintenance authorities (TMAs). To represent the system's structure and how different entities interact, a Class Diagram is designed.

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#### **Objective:**

To develop a UML Class Diagram that shows the static structure of the Career Guidance Portal and the association relationships among its main classes.

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#### **Definition:**

A Class Diagram in UML represents the system's structure by showing its classes and the relationships among them. It helps visualize how various entities interact and supports the design of databases and software components.

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#### **Main Classes and Relationships:**

- CareerGuidancePortal: Central hub connecting all users and services.
- Student: Registers, takes tests, and raises queries.
- Counsellor: Replies to student queries.
- Administrator: Approves registrations, advertisements, and papers.
- Advertiser: Submits and manages ads.
- TestMaintenanceAuthority: Uploads test papers for review.

- TestPaper: Created by TMA, attempted by students, approved by admin.
- Advertisement: Submitted by advertisers, approved by admin.
- Query: Raised by students and answered by counsellors.
- PaymentGateway & EmailService: Handle payments and notifications.

**Associations:**

- Student ↔ TestPaper (attempts)
  - TMA ↔ TestPaper (creates)
  - Student ↔ Query ↔ Counsellor (raises / answers)
  - Administrator ↔ TestPaper / Advertisement / Query (approves / monitors)
  - Portal ↔ PaymentGateway & EmailService (uses)
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**Conclusion:**

The Class Diagram clearly shows how different users and components interact within the Career Guidance Portal. It defines key entities and their associations, forming a solid base for database design and system implementation.